

ITMD 510: Object-Oriented App Develop

Java Cybersecurity Project – Swaha Enterprises



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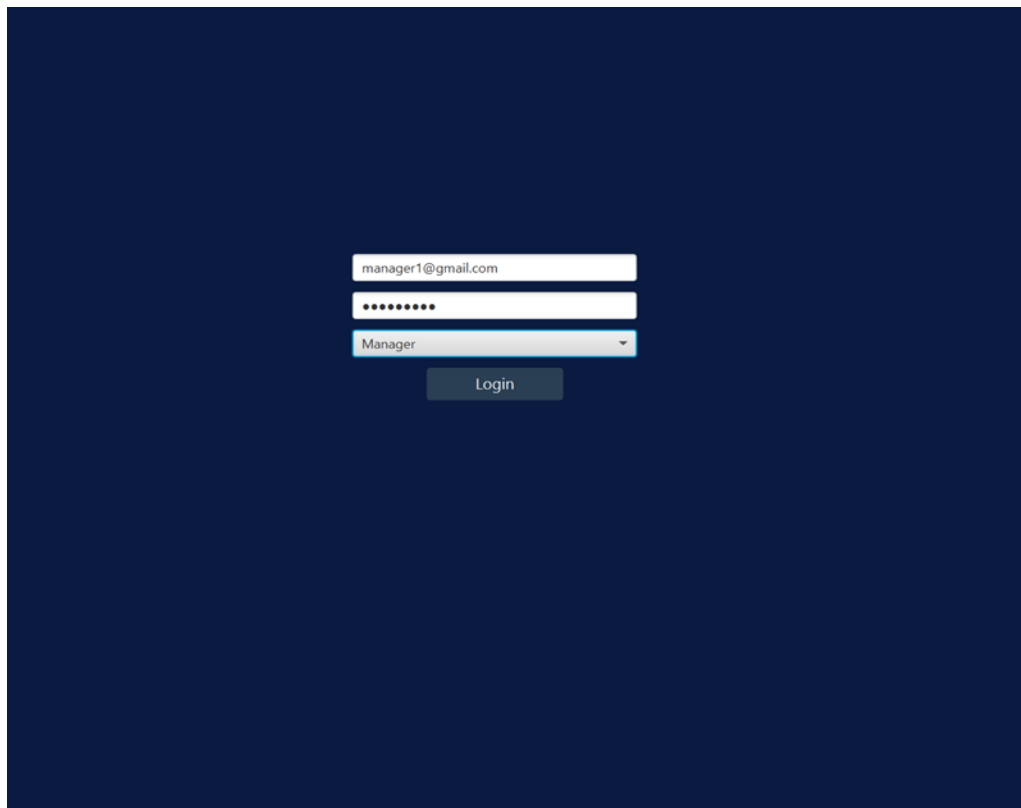
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1.Abstract:

This project presents a cybersecurity ticketing system designed to monitor and manage security-related incidents within an organization. The system focuses on detecting login failures, multiple unsuccessful authentication attempts, and unauthorized file access errors. The user hierarchy consists of three roles: Manager, Admin, and Employee. When an employee attempts to log in using incorrect credentials, accesses unauthorized resources, or fails authentication more than three times, the system automatically generates a security ticket. These tickets are handled by Admin users and can be reviewed by the Manager. The Manager has the authority to add or remove Admin users, while Admins are responsible for managing Employee accounts. This role-based structure ensures secure access control, accountability, and efficient handling of security incidents.

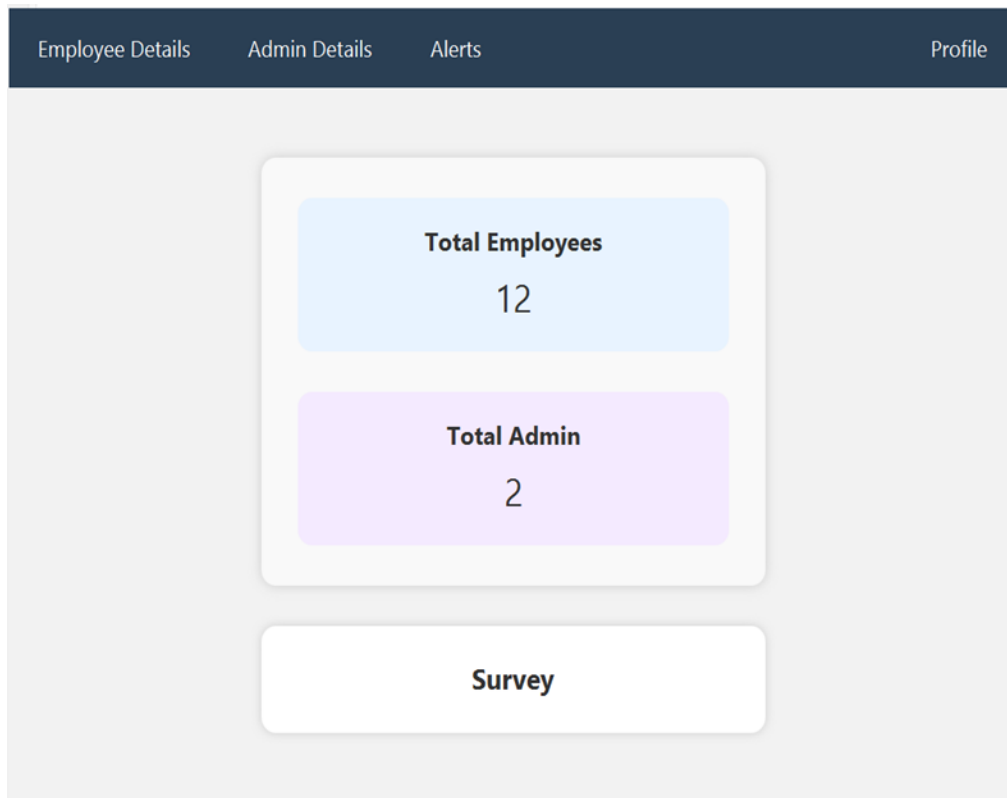
2.Login Page:

The login page serves as the central access point for all users, including Employees, Admins, and Managers, and redirects them to their respective dashboards upon successful authentication. If an employee enters an incorrect password more than three times or attempts to log in using unauthorized credentials, the system automatically generates a security ticket. For security purposes, all passwords are securely hashed during the authentication process.

A login form centered on a dark blue background. It consists of three white input fields stacked vertically. The first field contains the email address 'manager1@gmail.com'. The second field contains a series of dots, indicating a password. The third field is a dropdown menu with 'Manager' selected. Below these fields is a dark blue 'Login' button.

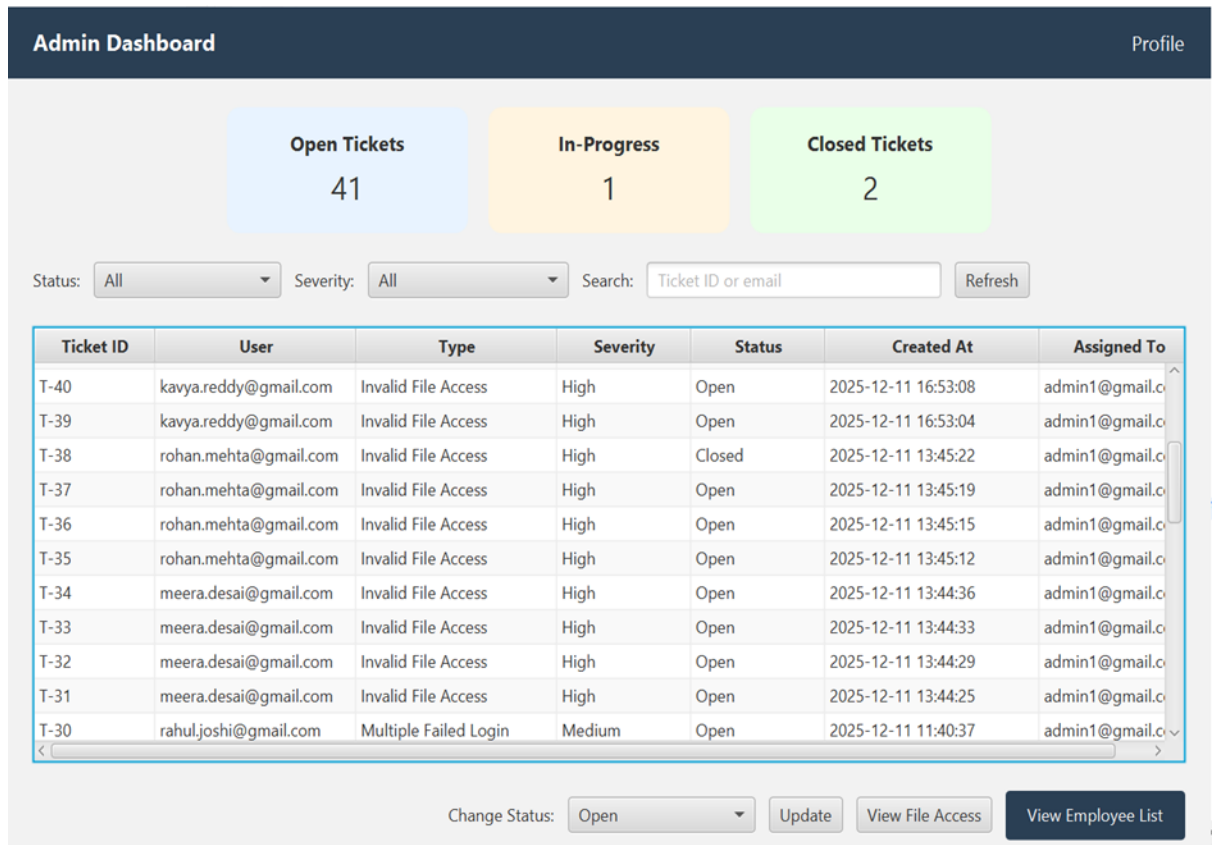
3.Manager Dashboard:

The Manager Dashboard provides an overview of the system, displaying the total number of employees and admins within the organization. It includes a survey section represented as a bar chart that visualizes previously recorded suspicious activities. The dashboard allows the manager to view detailed information about all employees and admins, as well as add or delete admin accounts. An alerts section enables the manager to review all generated security alerts. Under the profile section, the currently logged-in manager's details are displayed, along with an option to securely log out of the system.



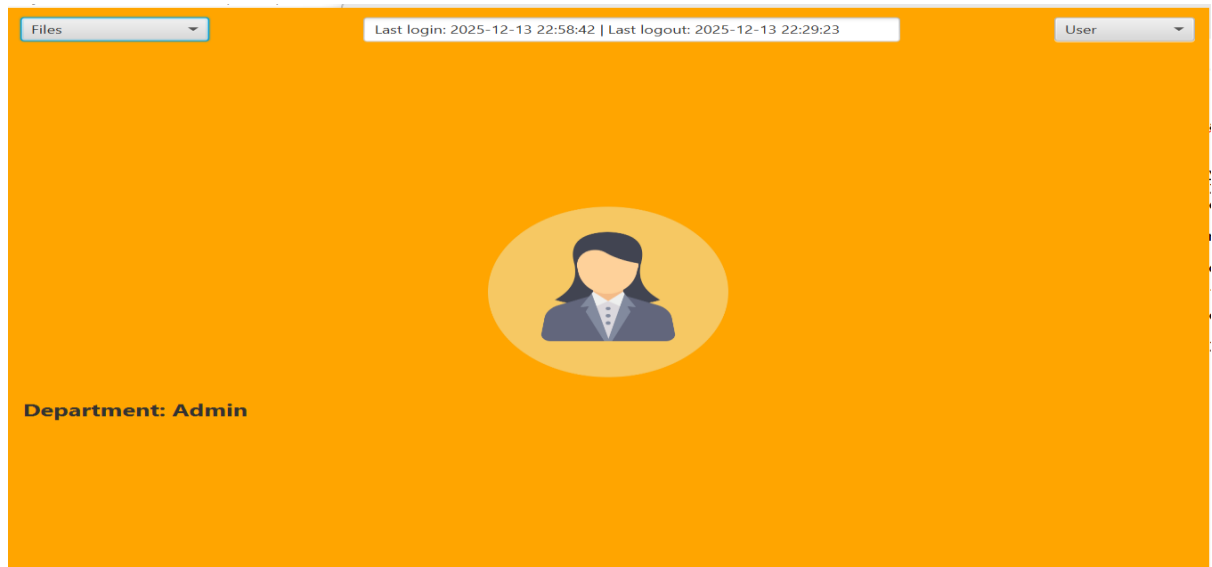
4.Admin Dashboard:

The Admin Dashboard allows administrators to monitor and manage suspicious activities through a ticket table that displays all generated security tickets. Admins can filter tickets by severity level and search using ticket IDs for quick access. At the bottom of the dashboard, admins can update the status of tickets to indicate whether they are in progress, being handled, or closed. Additionally, admins have the ability to view, add, and delete employee accounts. The profile section displays the admin's details and provides an option to securely log out.

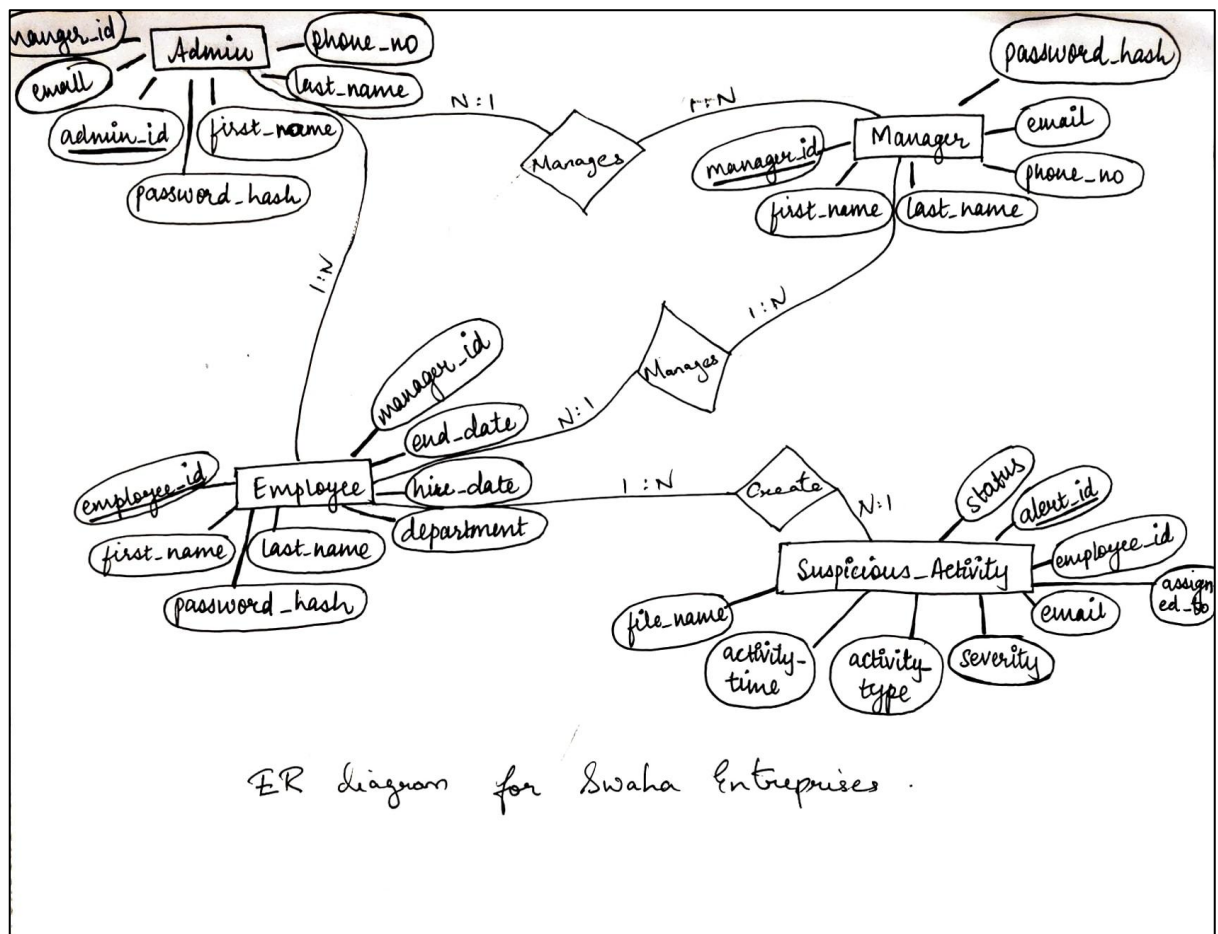


5.Employee Dashboard:

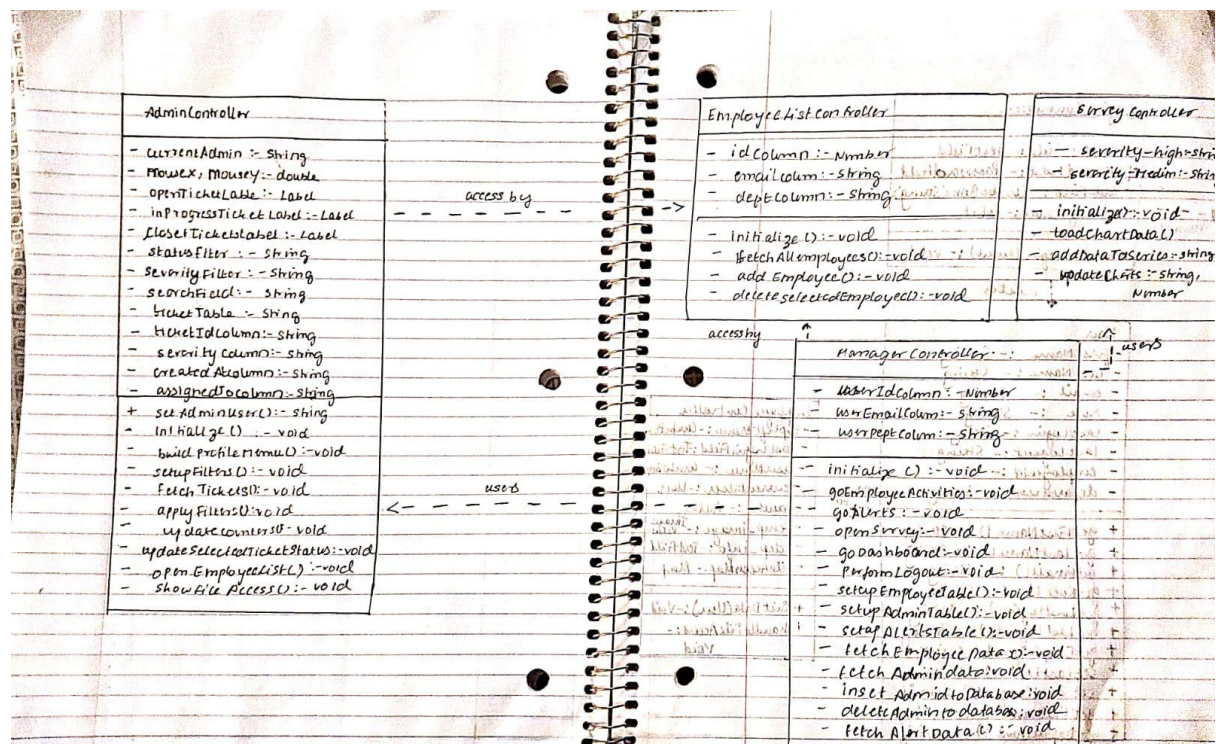
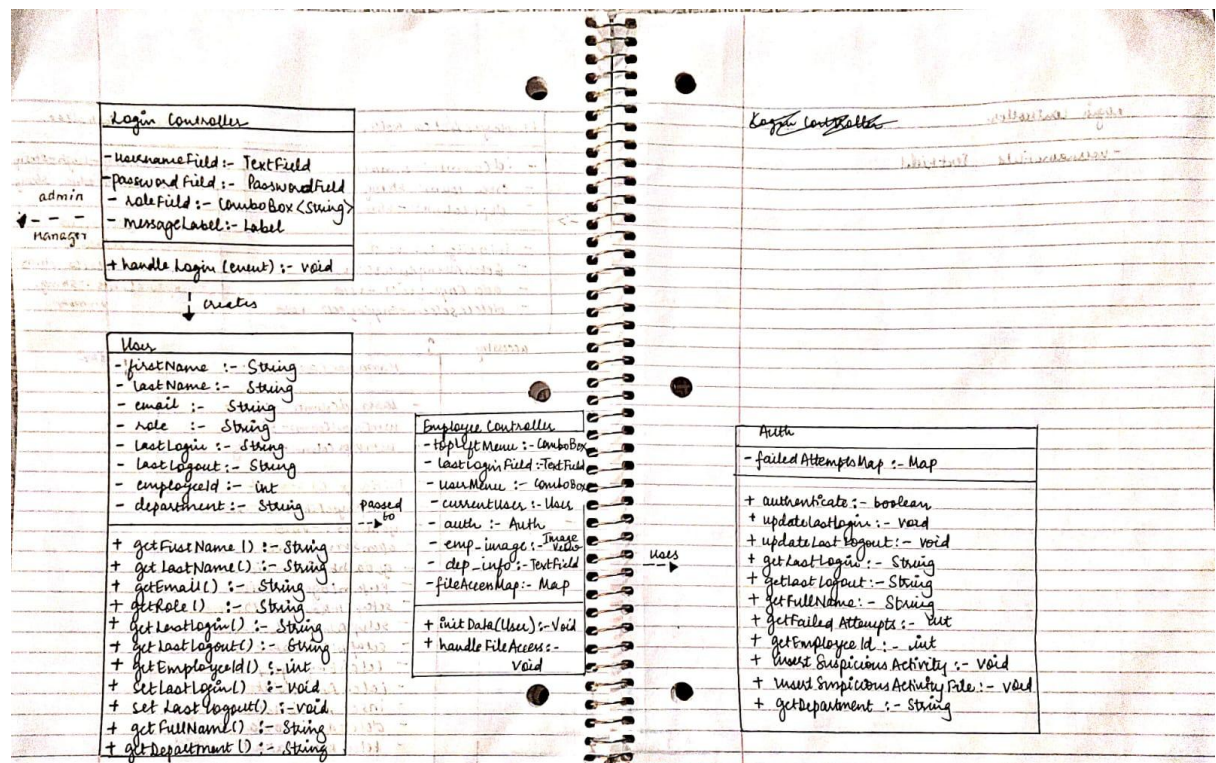
The Employee Dashboard displays the user's last login and last logout details. Employees can access authorized files, and if an employee attempts to access a file without the required permissions, an error message is displayed and the activity is recorded. Upon logging in, the dashboard shows the employee's department information. The profile section allows the employee to view their details and securely log out of the system.



6.ER Diagram:

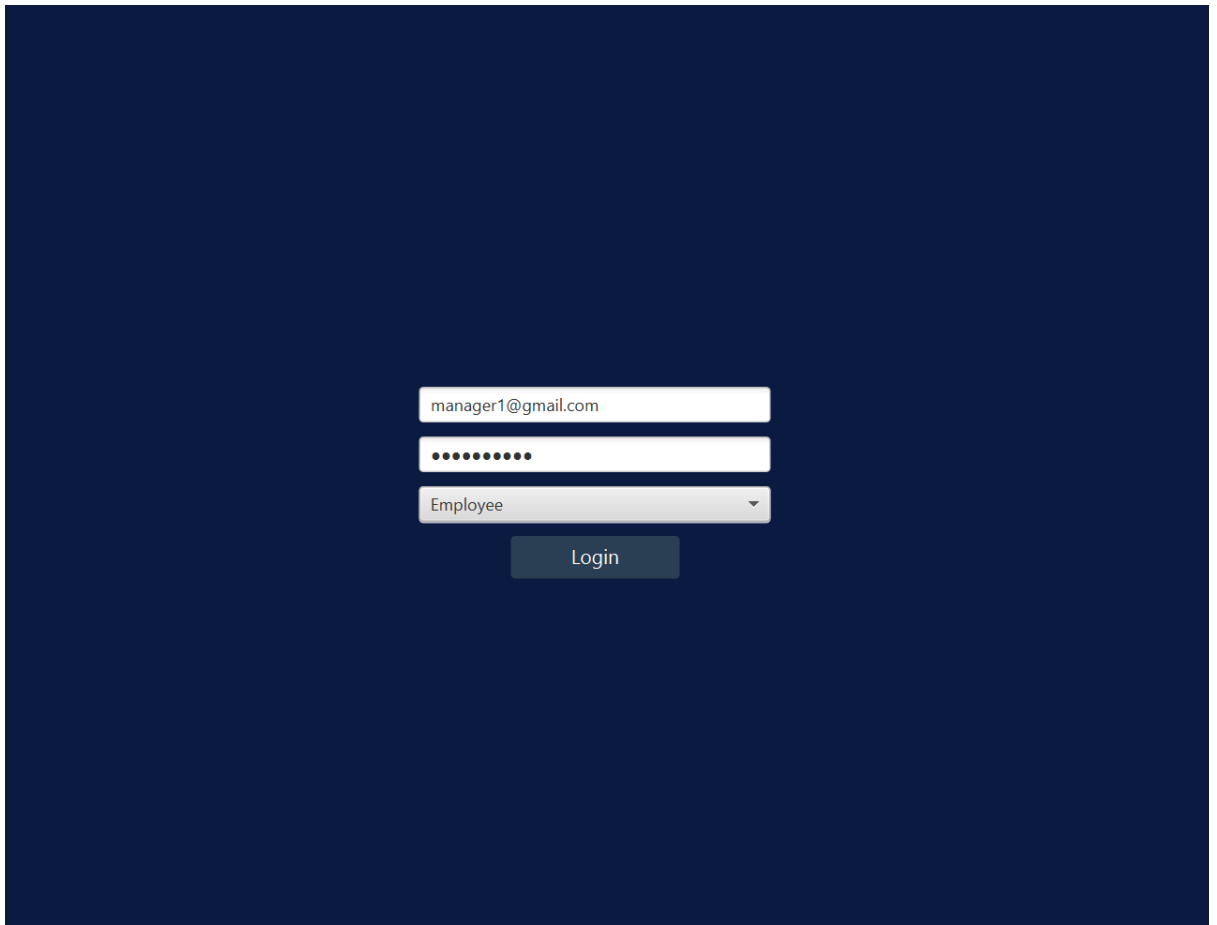


6. UML diagram:



7. Snapshots:

- **Successful Logins (Manager, admin, employee)**

A screenshot of a login interface on a dark blue background. The form is centered and consists of three input fields stacked vertically. The first field contains the email address 'manager1@gmail.com'. The second field contains a series of dots, indicating a password. The third field is a dropdown menu with 'Employee' selected. Below these fields is a dark blue 'Login' button.

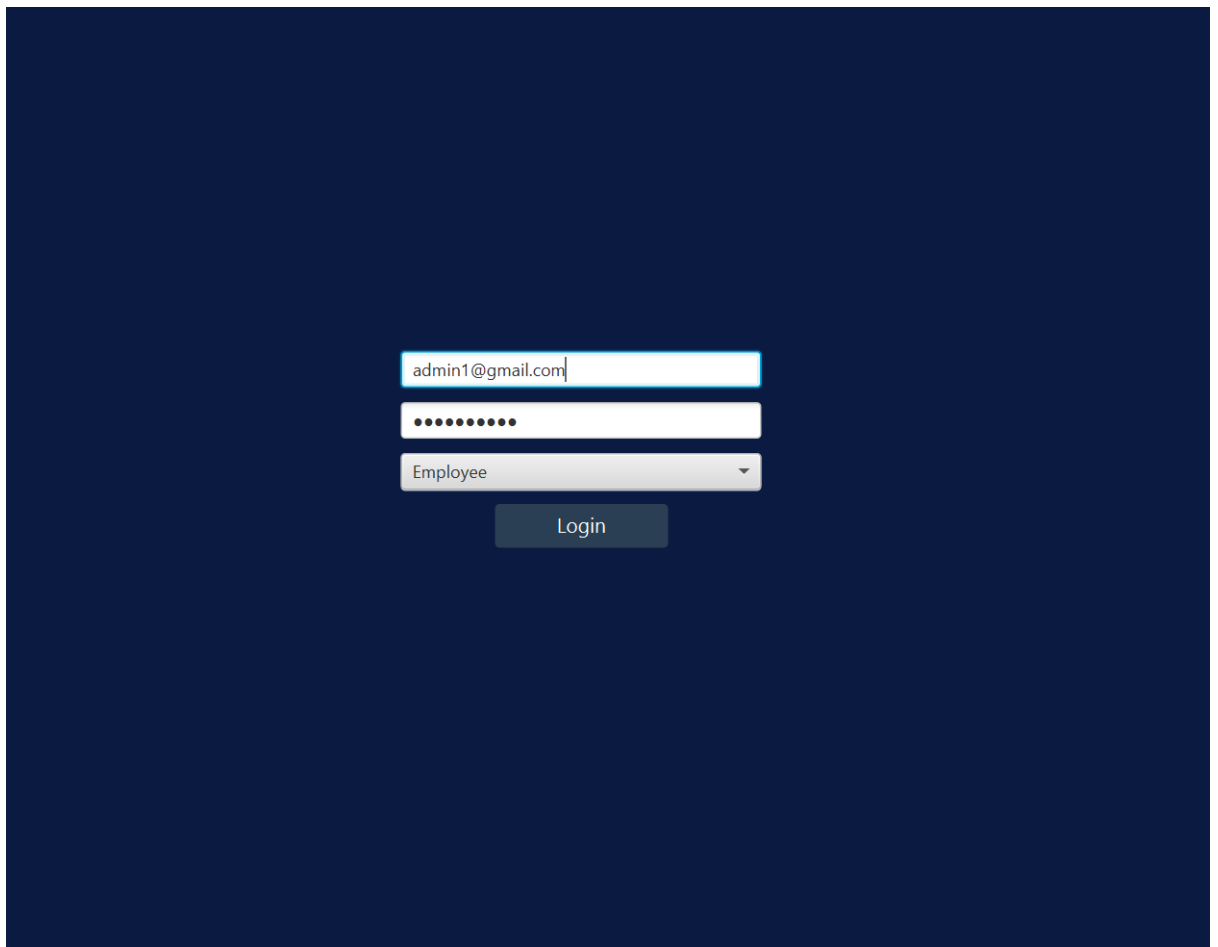
manager1@gmail.com

.....

Employee ▼

Login

Fig: ManagerLogin



The image shows a login interface on a dark blue background. At the top, there is a faint, light blue watermark that reads "Admin Login". Below this, centered on the page, is a login form. The form consists of three input fields stacked vertically: the first contains the email address "admin1@gmail.com", the second contains ten black dots representing a password, and the third is a dropdown menu currently showing "Employee". Below these fields is a dark blue button with the word "Login" in white text.

Fig: AdminLogin

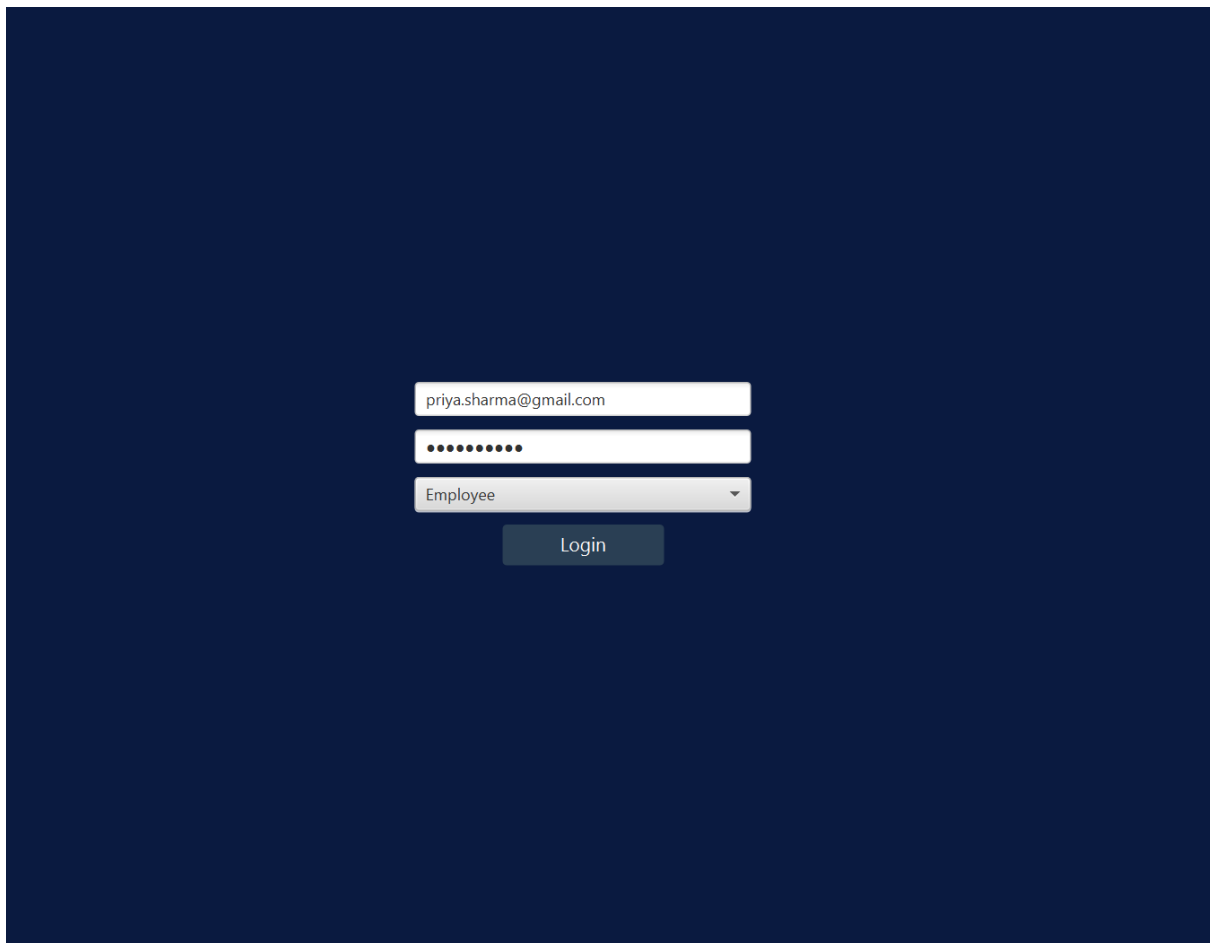


Fig : employee login

2) 10 records from database

The screenshot shows a database application interface. At the top, a toolbar includes icons for file operations and a 'Limit to 1000 rows' dropdown. Below the toolbar, a SQL editor contains the query '1 • select * from employee;'. To the right of the editor, a panel displays a message: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.' Below the SQL editor is a 'Result Grid' showing 10 records. The grid has columns for employee_id, first_name, last_name, email, department, and hire_date. The records are numbered 1 through 10. At the bottom of the interface, there is an 'Output' section with a table showing the execution of the query. The table has columns for #, Time, Action, Message, and Duration / Fetch. The first row shows a successful execution at 00:08:32, returning 12 rows in 0.032 seconds.

employee_id	first_name	last_name	email	department	hire_date
1	Rohan	Mehta	rohan.mehta@gmail.com	IT	2022-03-10
2	Priya	Sharma	priya.sharma@gmail.com	HR	2021-07-15
3	Arjun	Patel	arjun.patel@gmail.com	Finance	2023-01-20
4	Kavya	Reddy	kavya.reddy@gmail.com	Security	2022-11-05
5	Nikhil	Verma	nikhil.verma@gmail.com	IT	2020-09-01
6	Aisha	Khan	aisha.khan@gmail.com	Admin	2023-06-25
7	Rahul	Joshi	rahul.joshi@gmail.com	Operations	2021-02-18
8	Meera	Desai	meera.desai@gmail.com	Finance	2019-12-10
9	Vikram	Nair	vikram.nair@gmail.com	Security	2022-08-14
10	Tarun	Shetty	tarun.shetty@gmail.com	IT	2024-01-03

#	Time	Action	Message	Duration / Fetch
1	00:08:32	select * from employee LIMIT 0, 1000	12 row(s) returned	0.032 sec / 0.000 sec

3) Updates done by admin (before)

Admin Dashboard

Profile

Open Tickets44

In-Progress3

Closed Tickets2

Status:

All

Severity:

All

Search:

Ticket ID or email

Refresh

Ticket ID	User	Type	Severity	Status	Created At	Assigned To
T-49	rohan.mehta@gmail.com	Invalid File Access	High	Open	2025-12-13 23:40:57	
T-48	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:45:09	
T-47	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:37:56	
T-46	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:30:16	
T-45	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:30:06	
T-44	manager1@gmail.com	Login Failure	Medium	In-Progress	2025-12-13 16:30:04	admin1@gmail.co
T-43	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:29:55	
T-42	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:16	admin1@gmail.co
T-41	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:11	admin1@gmail.co
T-40	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:08	admin1@gmail.co
T-39	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:04	admin1@gmail.co

Change Status:

Open

Update

View File Access

View Employee List

Updates done by admin (after)

Admin Dashboard

Profile

Open Tickets43

In-Progress4

Closed Tickets2

Status:

All

Severity:

All

Search:

Ticket ID or email

Refresh

Ticket ID	User	Type	Severity	Status	Created At	Assigned To
T-49	rohan.mehta@gmail.com	Invalid File Access	High	Open	2025-12-13 23:40:57	
T-48	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:45:09	
T-47	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:37:56	
T-46	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:30:16	
T-45	manager1@gmail.com	Login Failure	Medium	In-Progress	2025-12-13 16:30:06	
T-44	manager1@gmail.com	Login Failure	Medium	In-Progress	2025-12-13 16:30:04	admin1@gmail.co
T-43	manager1@gmail.com	Login Failure	Medium	Open	2025-12-13 16:29:55	
T-42	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:16	admin1@gmail.co
T-41	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:11	admin1@gmail.co
T-40	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:08	admin1@gmail.co
T-39	kavya.reddy@gmail.com	Invalid File Access	High	Open	2025-12-11 16:53:04	admin1@gmail.co

Change Status:

In-Progress

Update

View File Access

View Employee List

4) Add & delete admin from manager

[illegible]

Fig : adding admin3

5) Popup alerts from employee file access

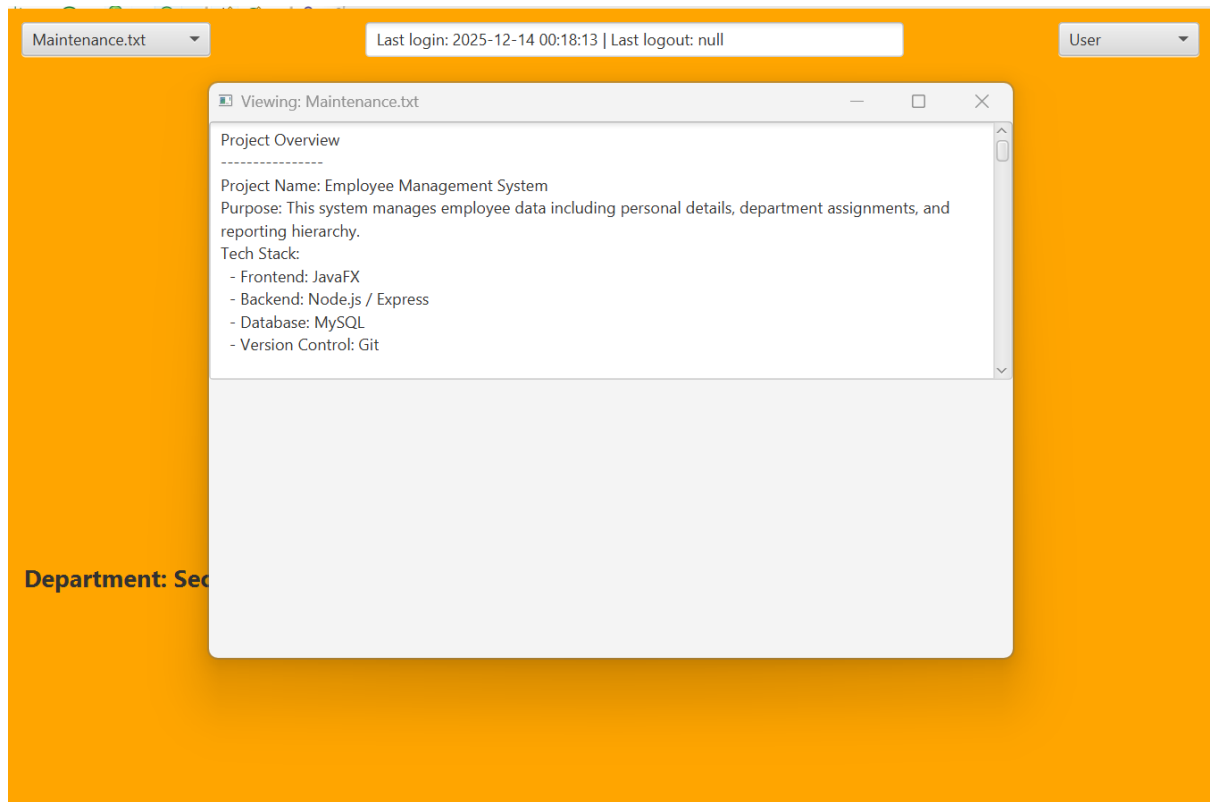


Fig: Vikram accessing maintenance file

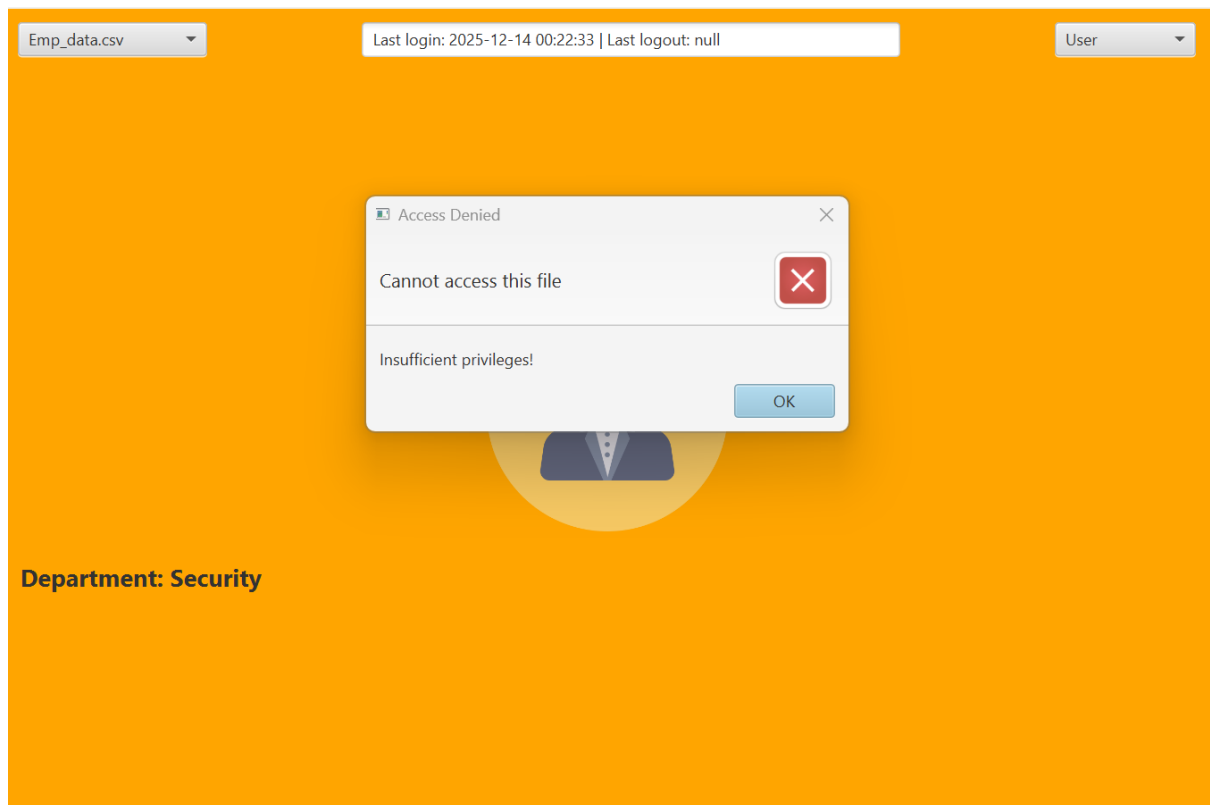


Fig: Vikram not able to access the file

7) Credentials for the users

Manager:

The screenshot shows the SQL File 3* interface. The SQL editor contains the query `select * from manager;`. The result grid displays the following data:

manager_id	first_name	last_name	phone_no	email	password_has
1	Sneha	Thakur	1234567890	manager1@gmail.com	manager_1
2	Swathi	Balachandar	3120987654	manager2@gmail.com	manager_2
NULL	NULL	NULL	NULL	NULL	NULL

The 'email' and 'password_has' columns are highlighted with a red box. The right sidebar contains a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

Admin:

The screenshot shows the SQL File 3* interface. The SQL editor contains the query `select * from admin;`. The result grid displays the following data:

admin_id	first_name	last_name	phone	email	manager_id	password_has
1	Aadhi	Krish	312...	admin1@gmail.com	1	admin_1
2	Jasmine	Patel	312...	admin2@gmail.com	2	admin_2
NULL	NULL	NULL	NULL	NULL	NULL	NULL

The 'email' and 'password_has' columns are highlighted with a red box. The right sidebar contains a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom section shows the Action Output log:

#	Time	Action	Message	Duration / Fetch
2	00:24:16	select * from employee LIMIT 0, 1000	12 row(s) returned	0.047 sec / 0.000 sec
3	00:25:30	select * from manager LIMIT 0, 1000	2 row(s) returned	0.031 sec / 0.000 sec

Employee:

SQL File 3* x

Limit to 1000 rows

1 select * from employee;

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

email	department	hire_date	end_date	mar	password_hash
rohan.mehta@gmail.com	IT	2022-03-10	NULL	1	employee_1
priya.sharma@gmail.com	HR	2021-07-15	NULL	2	employee_2
arjun.patel@gmail.com	Finance	2023-01-20	NULL	1	employee_3
kavya.reddy@gmail.com	Security	2022-11-05	NULL	2	employee_4
nikhil.verma@gmail.com	IT	2020-09-01	2024-01-15	1	employee_5
aisha.khan@gmail.com	Admin	2023-06-25	NULL	2	employee_6
rahul.joshi@gmail.com	Operations	2021-02-18	NULL	1	employee_7
meera.desai@gmail.com	Finance	2019-12-10	NULL	2	employee_8
vikram.nair@gmail.com	Security	2022-08-14	NULL	1	employee_9
tarun.shetty@gmail.com	IT	2024-01-03	NULL	2	employee_10

employee 2 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	00:08:32	select * from employee LIMIT 0, 1000	12 row(s) returned	0.032 sec / 0.000 sec
2	00:24:16	select * from employee LIMIT 0, 1000	12 row(s) returned	0.047 sec / 0.000 sec

GitHub link :

<https://github.com/Swaadhi/FinalProjectFall2025.git>